

Problem 1:

Matlab Code:

% Reversible liquid-phase isomerization in CSTR

clear; clc;

% Constants

R = 1.987; % cal/mol-K

dH = -25000; % cal/mol (exothermic)

E = 15000; % cal/mol

Kc0 = 3.0; % equilibrium constant at 300 K

T0 = 300; % reference T, K

% Step 1: Find alpha from experimental data at 300 K

X_exp = 0.40; % given conversion at 300 K

f_alpha = @(alpha) alpha*((1-X_exp)^2 - (1/Kc0)*X_exp^2) - X_exp;

alpha0 = fzero(f_alpha, 1); % solve for alpha at 300 K

fprintf('Alpha at 300 K = %.4f\n', alpha0);

% Step 2: Define functions for Kc(T) and alpha(T)

Kc = @(T) Kc0*exp((dH/R)*(1/T0 - 1./T));

alpha = @(T) alpha0*exp(-E/R*(1./T - 1/T0));

% Step 3: Loop over T and solve for X(T)

Tvals = linspace(280, 340, 200); % temperature range (K)

Xvals = zeros(size(Tvals));

for i = 1:length(Tvals)

 Ti = Tvals(i);

 Kci = Kc(Ti);

 alphai = alpha(Ti);

 % Define the nonlinear equation in X

 funX = @(X) alphai*((1-X).^2 - (1/Kci)*X.^2) - X;

 % Solve with initial guess near 0.4

 Xi = fzero(funX, 0.4);

 Xvals(i) = Xi;

end

% Step 4: Find maximum conversion and corresponding T

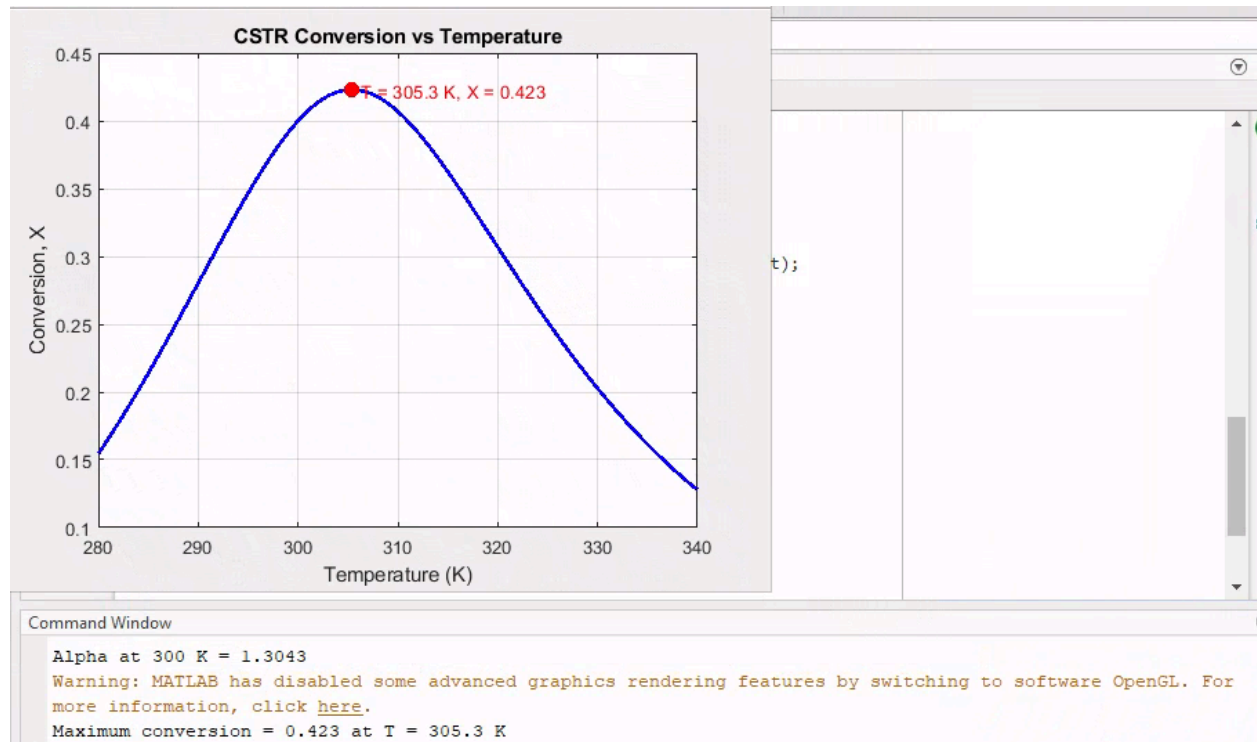
[Xmax, idx] = max(Xvals);

Topt = Tvals(idx);

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% Step 5: Plot
figure;
plot(Tvals, Xvals, 'b-', 'LineWidth', 2); hold on;
plot(Topt, Xmax, 'ro', 'MarkerSize', 8, 'MarkerFaceColor', 'r'); % optimum point
text(Topt+1, Xmax, sprintf('T = %.1f K, X = %.3f', Topt, Xmax), ...
     'FontSize', 10, 'Color', 'r');

xlabel('Temperature (K)');
ylabel('Conversion, X');
title('CSTR Conversion vs Temperature');
grid on;

fprintf('Maximum conversion = %.3f at T = %.1f K\n', Xmax, Topt);
```



I recommend the CSTR to be run at 305.3 K because the maximum conversion happens at this point 42.3% conversion.

Problem 2:

c) $t = \frac{q}{k} : K(273) = 0.00254 \text{ dm}^3/\text{mol} \cdot \text{min}$

$t = \frac{q}{0.00254} = 3550.2 \text{ min} = 59.17 \text{ hr} = 2.5 \text{ days}$

d)

Series: CSTR $\rightarrow$ PFR

$x = 0.4396, C_{A, \text{near core}} = 1(1-x) = 0.5604 \text{ M}$

$\frac{1}{C_{\text{out}}} = \frac{1}{0.5604} + \frac{0.07 \cdot 800}{1} \Rightarrow C_{\text{out}} = 0.1354 \text{ M}$

$C_0 = 1; X_{\text{overall}} = 1 - 0.1354 = 0.8646, 86.46\% \text{ conversion}$

Parallel:

Split feed, $F = 5 \text{ mol/min}, C = 1 \text{ M}$

PFR: $\theta = \frac{V_R}{F} = \frac{800 \cdot 0.07}{5} = 11.2, X_{\text{PFR}} = \frac{11.2}{12.2} = 0.9180, 91.8\%$

CSTR: $5x = 200 \cdot 0.07(1-x), X_{\text{CSTR}} = 0.555$

$X_{\text{overall}} = (5 \cdot 0.9180) + (5 \cdot 0.555) / 10 = 0.7364, 73.64\%$

e)

$r = 10 \text{ dm}^3/\text{min}, C_A = 1 \text{ mol/dm}^3, X = 0.9$

$\dot{n}_A = C_{A0} r X = 10 \cdot 1 \cdot 0.9 = 9 \text{ mol/min} = 12,960 \text{ mol/day}$

Batch reactor:

$t = \int_{C_A}^{C_{A0}} \frac{dC_A}{-r_A} = \frac{1}{k} \left(\frac{1}{C_A} - \frac{1}{C_{A0}} \right) = \frac{1}{0.07} \left(\frac{1}{0.1} - \frac{1}{1} \right) = 128.57 \text{ min}$

mol A per batch = $C_{A0} V_{BR} X = 0.9 \cdot V_{BR}$

batches = $\frac{12,960}{0.9 \cdot V_{BR}}$

Max batches/day = $\frac{12,960}{150} = 86.4 \approx 86$

$q = \frac{12,960}{0.9 \cdot V_{BR}} \Rightarrow 9 \cdot 0.9 \cdot V_{BR} = 12,960 \Rightarrow V_{BR} = \frac{12,960}{8.1} = 1600 \text{ L} = 1600 \text{ dm}^3$

Table 1.1 - 500 gal = 190,000

f) The point of the problem is to emphasize the points that reactor type and operating temperature change conversion heavily and that design is a tradeoff between space and energy.

g) ways to do this problem incorrectly is forgetting the 1:1 dilution on mixing (using $C_0 = 2 \text{ M}$ instead of 1 M), using the wrong order rate law, ignoring temperature dependence of k

2. $A+B \rightarrow C$ $r = k C_A C_B$ $A=2M, B=2M, v_0=10 \text{ dm}^3/\text{min}, C_{A0}=C_{B0}=1M$
 $K(310) = 0.07 \text{ dm}^3/\text{mol} \cdot \text{min}, E = 20 \text{ kJ/mol}, R = 0.001987 \text{ kJ/mol} \cdot K$
 (STR) $V_{STR} = 200 \text{ dm}^3$ (can be heated to 350 K or cooled to 273 K)
 PFR $V_{PFR} = 800 \text{ dm}^3$ at fixed 300 K

a)

PFR: $V = \frac{F_{A0} X}{k C_{A0}^2 (1-X)^2}$ $F_{A0} = C_{A0} v_0 = 10 \text{ mol/min}$
 $\theta = \frac{V}{F_{A0}} = \frac{800}{10} = 80$
 $\theta = \frac{800 \cdot 0.7}{10} = 56$ $X_{PFR} = \frac{56}{66} = 0.8485, 84.85\%$

$K(300) = 0.07 \text{ dm}^3/\text{mol} \cdot \text{min}$
 $K(T) = 0.07 e^{\left(-\frac{E}{R} \left(\frac{1}{T} - \frac{1}{300}\right)\right)} = 0.07 e^{\left(-\frac{20}{0.001987} \left(\frac{1}{T} - \frac{1}{300}\right)\right)}$

$K(350) = 8.443 \text{ dm}^3/\text{mol} \cdot \text{min}$

$K(273) = 0.00254 \text{ dm}^3/\text{mol} \cdot \text{min}$

CSTR: $F_{A0} X = V k [C_{A0}(1-X)]^2$

at 300 K $10X = \frac{200(0.07)}{(1-X)^2} \Rightarrow X_{CSTR} = 0.4396, 43.96\%$

at 350 K $10X = 200(8.443)(1-X)^2 \Rightarrow X_{CSTR} = 0.9259, 92.59\%$

If the heating cost is acceptable, you should heat the CSTR to 350 K due to its small volume (doesn't take up space) and it has a high conversion, if the weather conditions are beneficial, it may also lower heating costs. If you want operation with no heating and huge space for the larger reactor, it gives 85% conversions without heating costs. If it is very cold weather, and there is no heat, the PFR would be better than the CSTR.

b) $X = 0.9, C = 0.1M$

$\frac{dC}{dt} = -kC^2$ $\text{limit } t \rightarrow \frac{1}{k} \left(\frac{1}{C} - \frac{1}{C_0}\right) = \frac{a}{k}$

$\frac{1}{C} - \frac{1}{C_0} = kt$ $K(350) = 8.443 \text{ dm}^3/\text{mol} \cdot \text{min}$

$t = \frac{\frac{1}{0.1} - \frac{1}{1}}{8.443} = 1.07 \text{ min}$